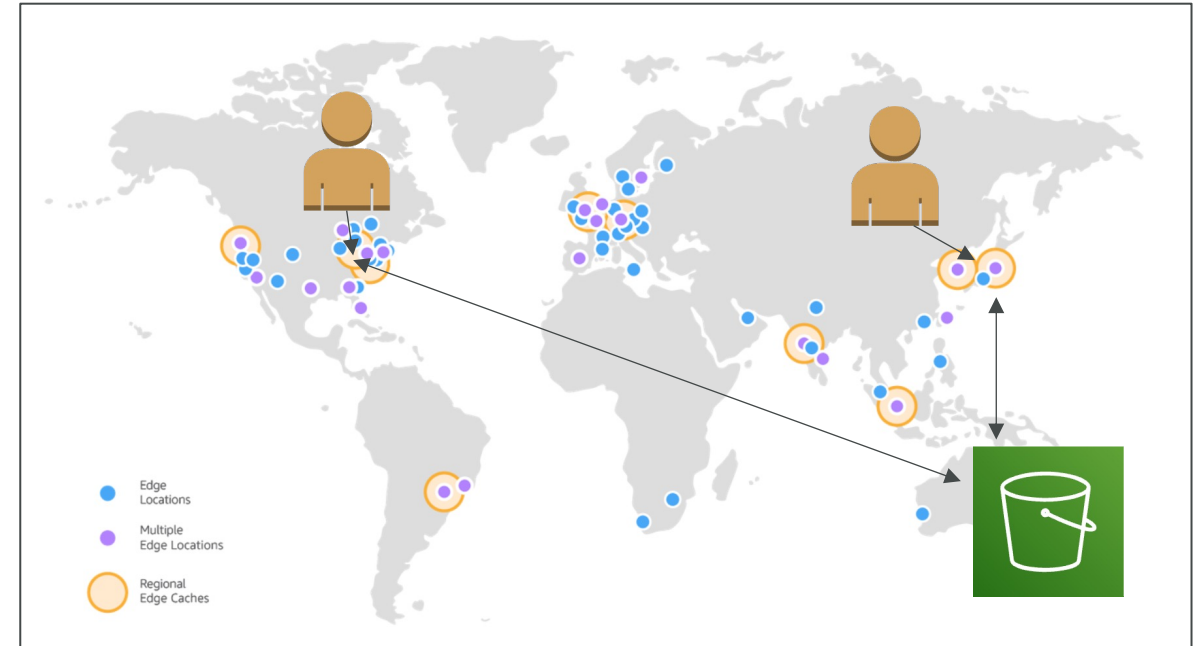


Amazon CloudFront



Amazon CloudFront

- Content Delivery Network (CDN)
- Improves read performance, content is cached at the edge
- Improves users experience
- Hundreds of Points of Presence globally (edge locations, caches)
- DDoS protection (because worldwide), integration with Shield, AWS Web Application Firewall

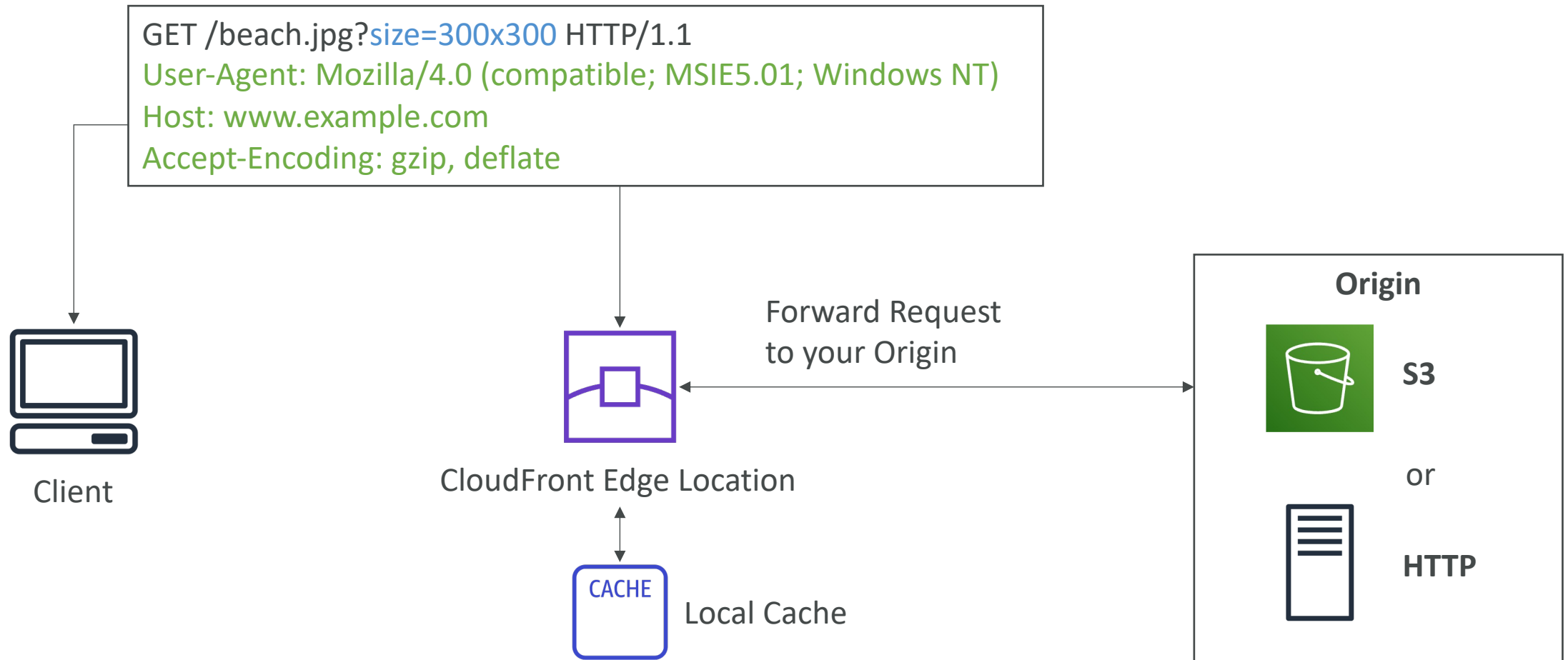


Source: <https://aws.amazon.com/cloudfront/features/?nc=sn&loc=2>

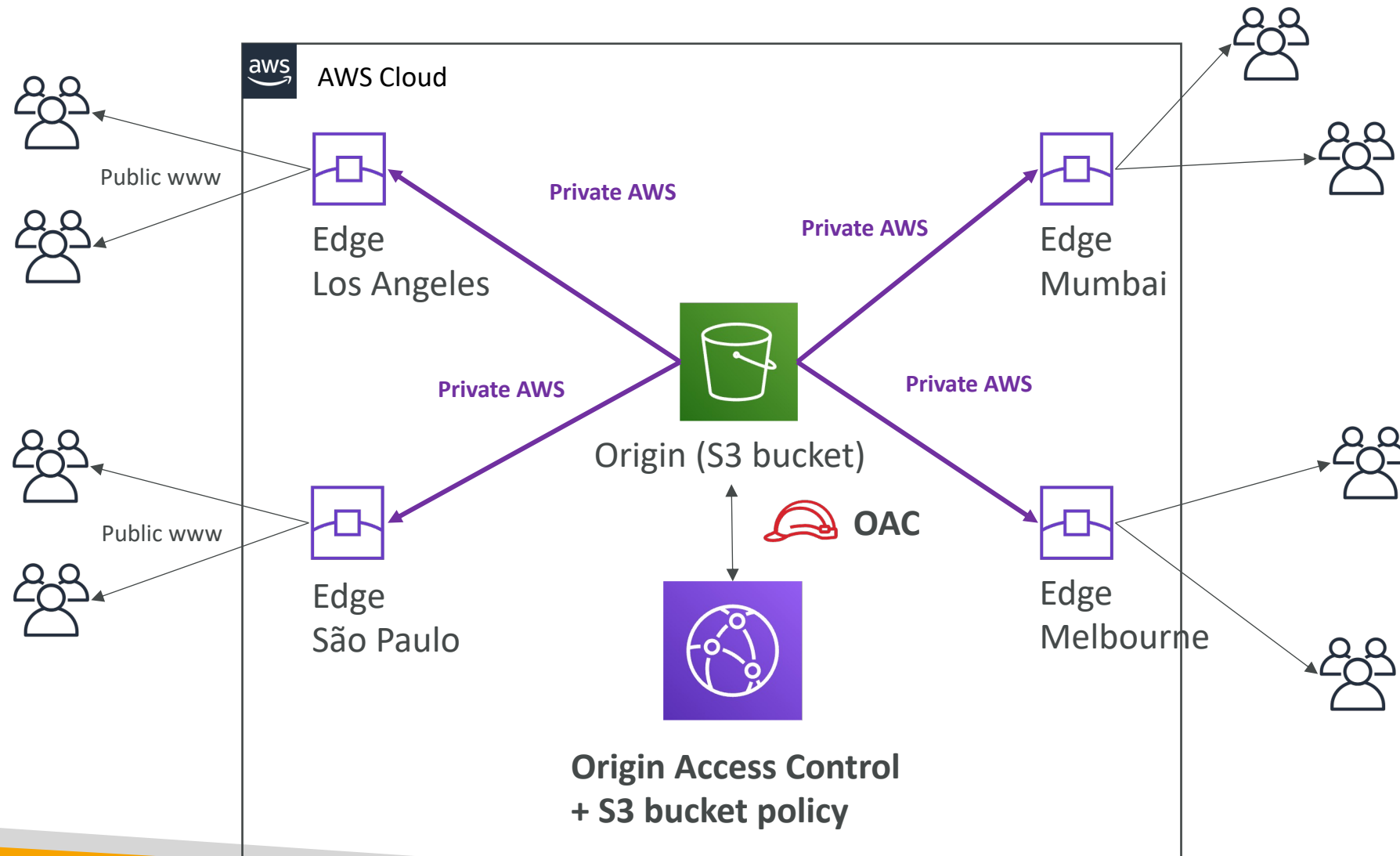
CloudFront – Origins

- S3 bucket
 - For distributing files and caching them at the edge
 - For uploading files to S3 through CloudFront
 - Secured using Origin Access Control (OAC)
- VPC Origin
 - For applications hosted in VPC private subnets
 - Private Application Load Balancer / Network Load Balancer / EC2 Instances
- Custom Origin (HTTP)
 - S3 website (must first enable the bucket as a static S3 website)
 - Any public HTTP backend you want (example: Public ALB)

CloudFront at a high level



CloudFront – S3 as an Origin

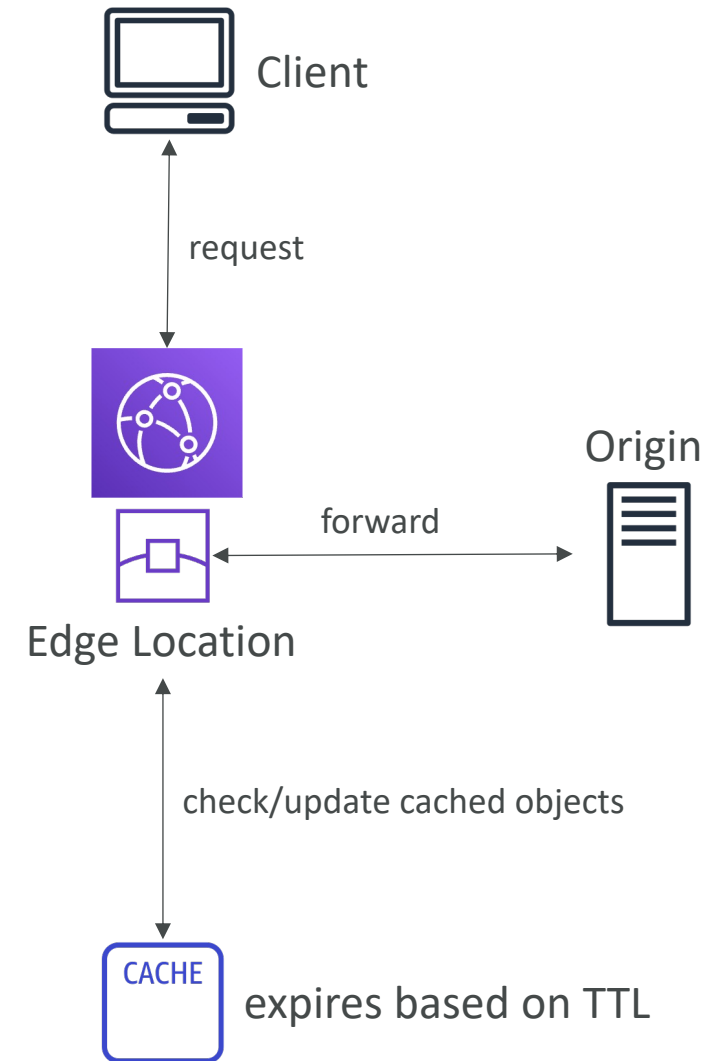


CloudFront vs S3 Cross Region Replication

- CloudFront:
 - Global Edge network
 - Files are cached for a TTL (maybe a day)
 - Great for static content that must be available everywhere
- S3 Cross Region Replication:
 - Must be setup for each region you want replication to happen
 - Files are updated in near real-time
 - Read only
 - Great for dynamic content that needs to be available at low-latency in few regions

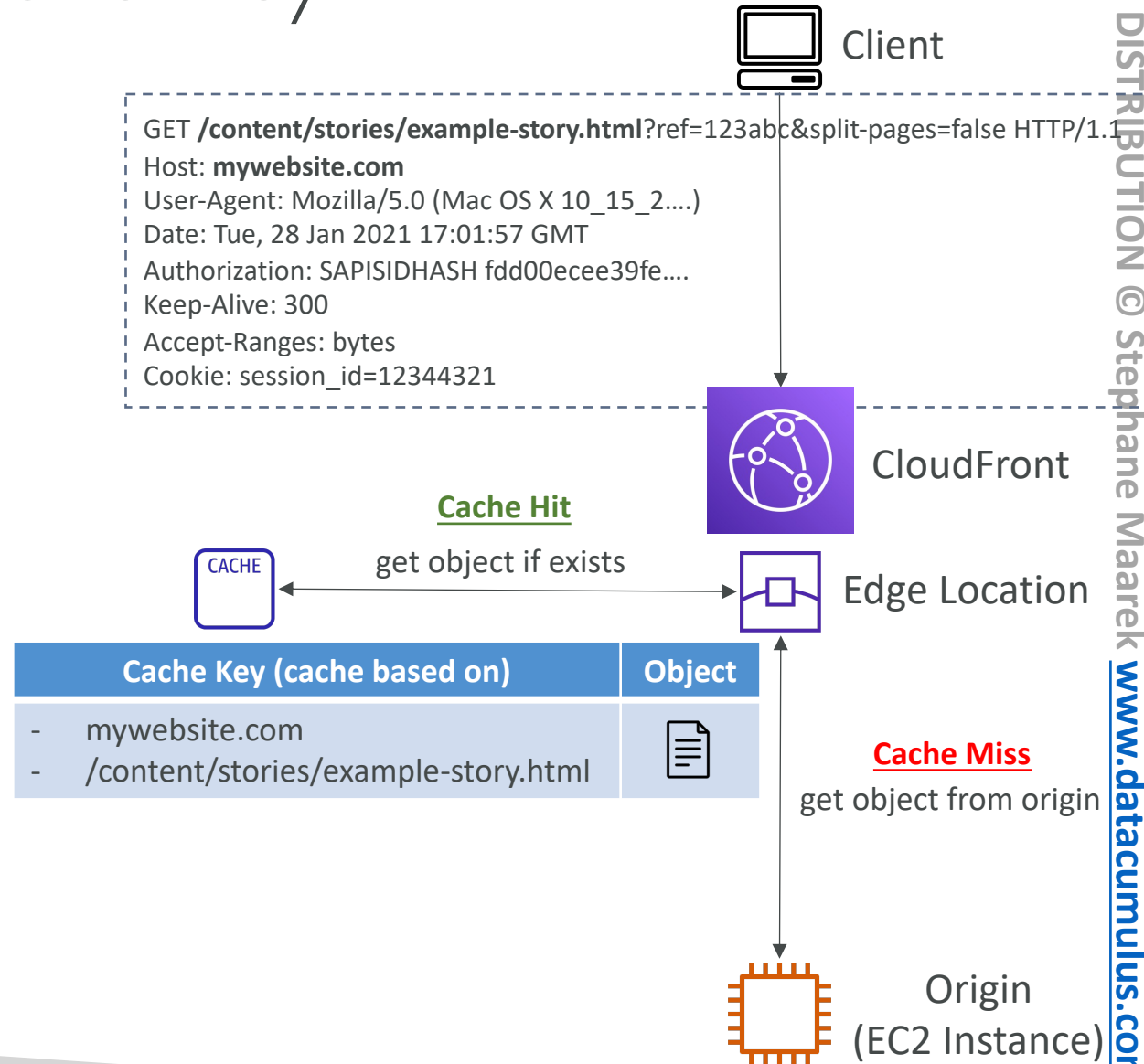
CloudFront Caching

- The cache lives at each CloudFront **Edge Location**
- CloudFront identifies each object in the cache using the **Cache Key** (see next slide)
- You want to maximize the Cache Hit ratio to minimize requests to the origin
- You can invalidate part of the cache using the **CreateInvalidation API**



What is CloudFront Cache Key?

- A unique identifier for every object in the cache
- By default, consists of **hostname + resource portion of the URL**
- If you have an application that serves up content that varies based on user, device, language, location...
- You can add other elements (HTTP headers, cookies, query strings) to the Cache Key using **CloudFront Cache Policies**



CloudFront Policies – Cache Policy

- Cache based on:
 - **HTTP Headers:** None – Whitelist
 - **Cookies:** None – Whitelist – Include All-Except – All
 - **Query Strings:** None – Whitelist – Include All-Except – All
- Control the TTL (0 seconds to 1 year), can be set by the origin using the **Cache-Control** header, **Expires** header...
- Create your own policy or use Predefined Managed Policies
- All HTTP headers, cookies, and query strings that you include in the Cache Key are automatically included in origin requests

CloudFront Caching – Cache Policy HTTP Headers

GET /blogs/myblog.html HTTP/1.1
Host: mywebsite.com
User-Agent: Mozilla/5.0 (Mac OS X 10_15_2....)
Date: Tue, 28 Jan 2021 17:01:57 GMT
Authorization: SAPISIDHASH fdd00ecee39fe....
Keep-Alive: 300
Language: fr-fr



Client

request



CloudFront

- **None:**
 - Don't include any headers in the Cache Key (except default)
 - Headers are not forwarded (except default)
 - Best caching performance
- **Whitelist:**
 - **only specified headers** included in the Cache Key
 - Specified headers are also forwarded to Origin

CloudFront Cache – Cache Policy Query Strings

GET /image/cat.jpg?border=red&size=large HTTP/1.1

...



Client

request



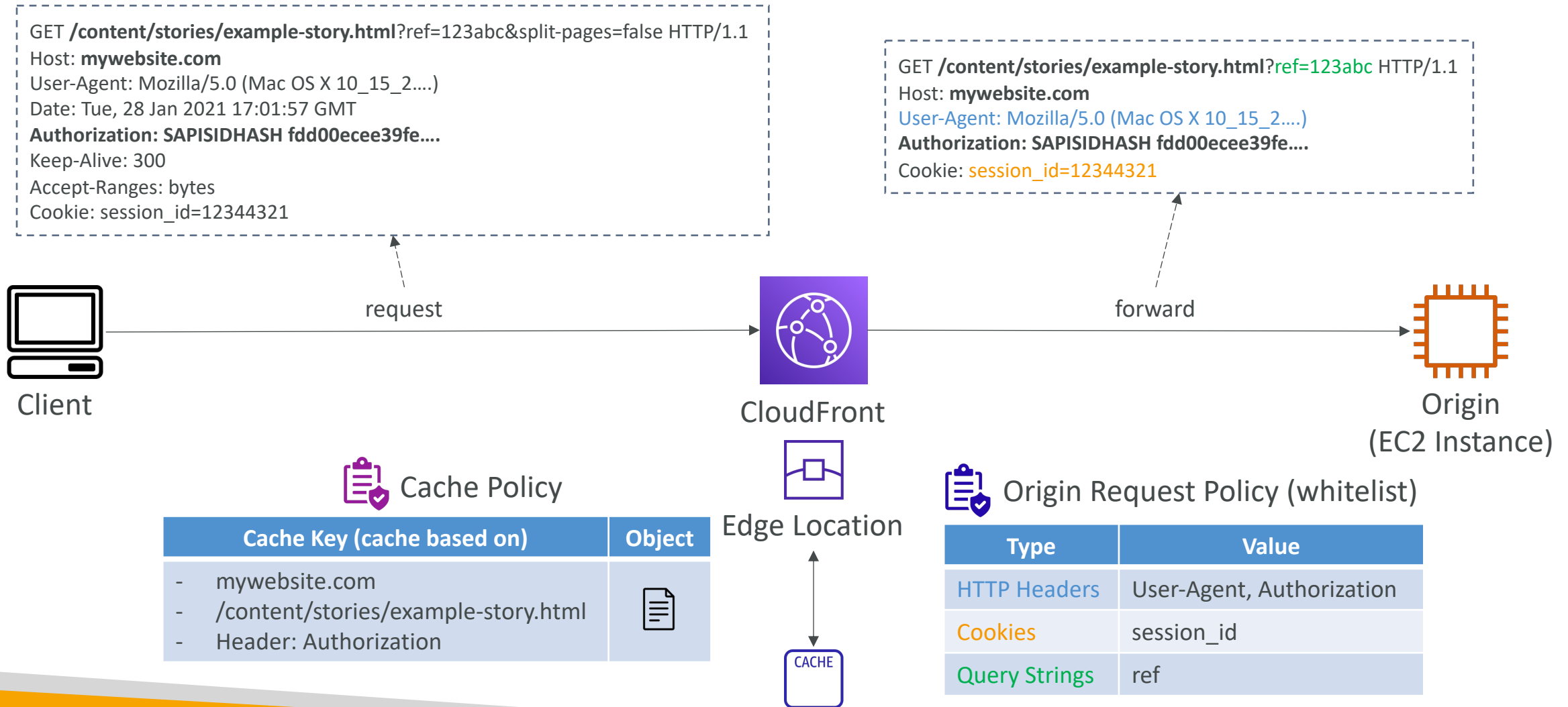
CloudFront

- **None**
 - Don't include any query strings in the Cache Key
 - Query strings are not forwarded
- **Whitelist**
 - Only specified query strings included in the Cache Key
 - Only specified query strings are forwarded
- **Include All-Except**
 - Include all query strings in the Cache Key except the specified list
 - All query strings are forwarded except the specified list
- **All**
 - Include all query strings in the Cache Key
 - All query strings are forwarded
 - Worst caching performance

CloudFront Policies – Origin Request Policy

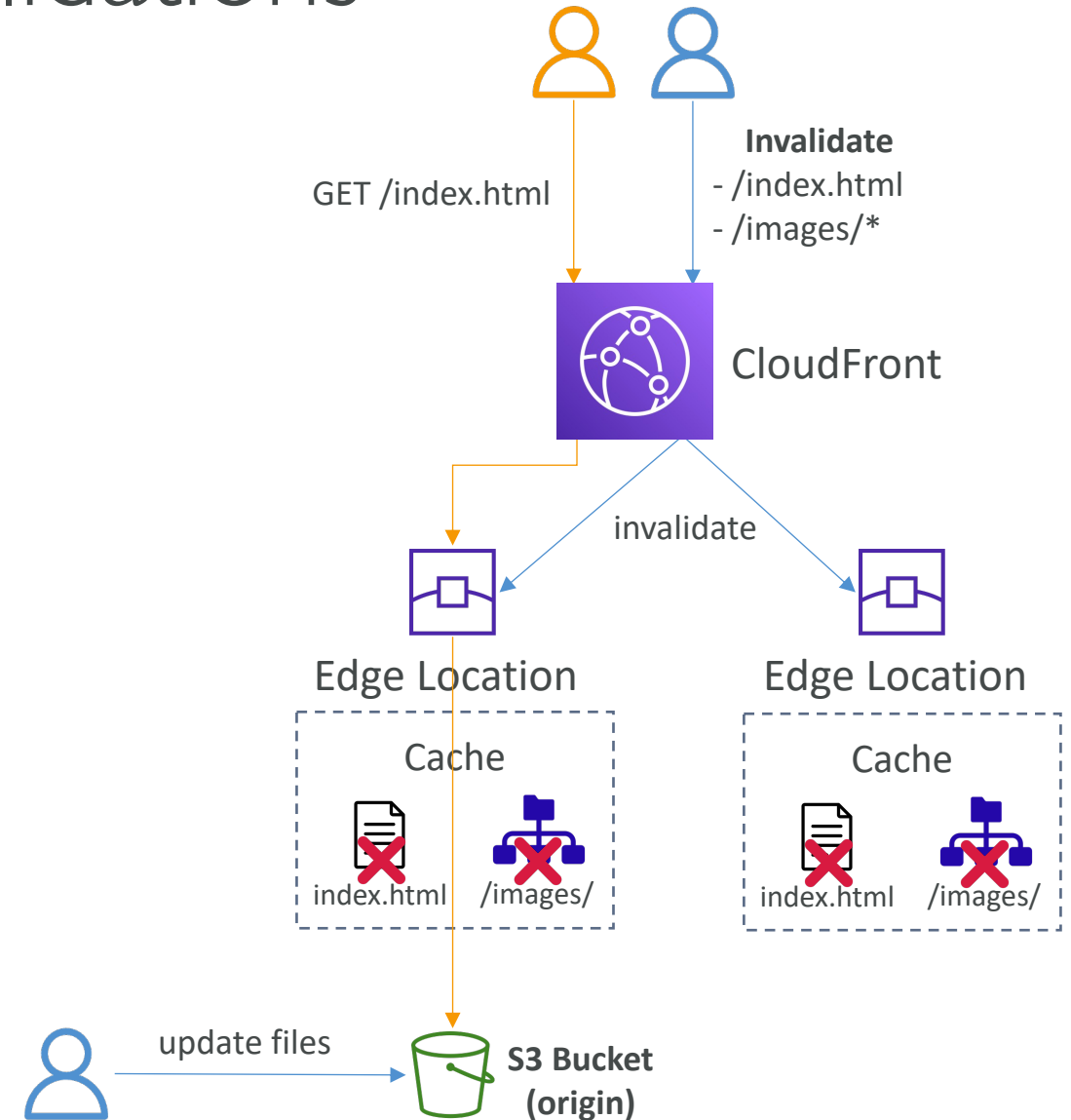
- Specify values that you want to include in origin requests **without** including them in the Cache Key (no duplicated cached content)
- You can include:
 - **HTTP headers:** None – Whitelist – All viewer headers options
 - **Cookies:** None – Whitelist – All
 - **Query Strings:** None – Whitelist – All
- Ability to add CloudFront HTTP headers and Custom Headers to an origin request that were not included in the viewer request
- Create your own policy or use Predefined Managed Policies

Cache Policy vs. Origin Request Policy



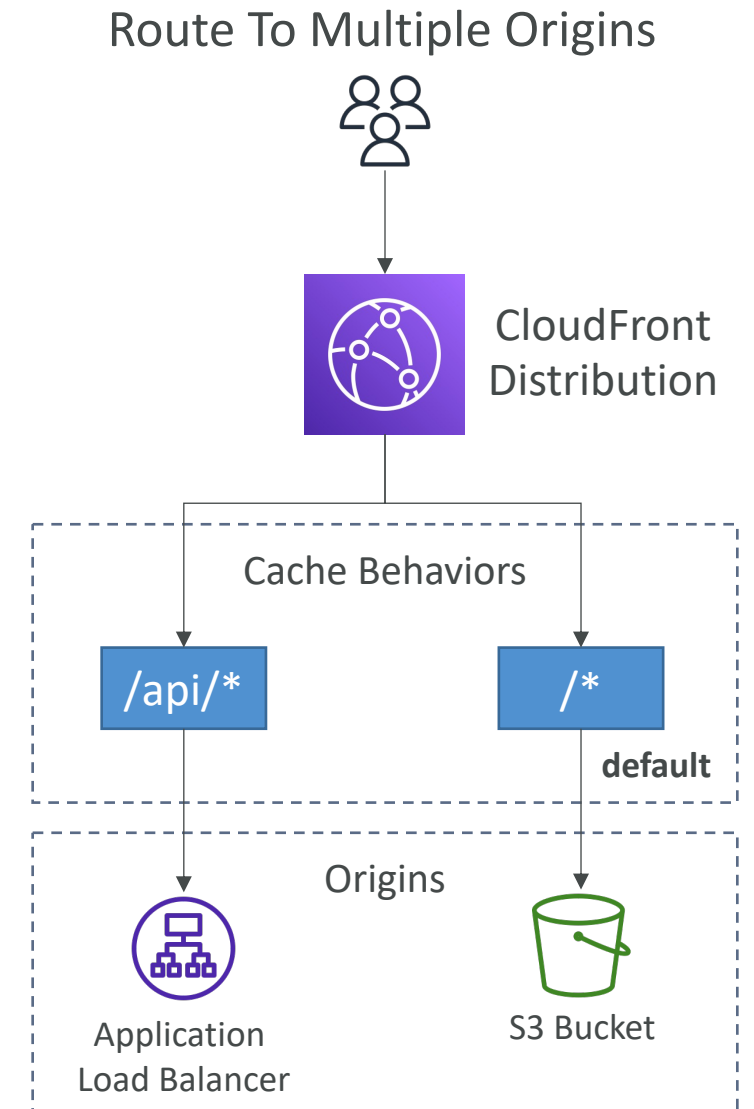
CloudFront – Cache Invalidations

- In case you update the back-end origin, CloudFront doesn't know about it and will only get the refreshed content after the TTL has expired
- However, you can force an entire or partial cache refresh (thus bypassing the TTL) by performing a **CloudFront Invalidation**
- You can invalidate all files (*) or a special path (/images/*)

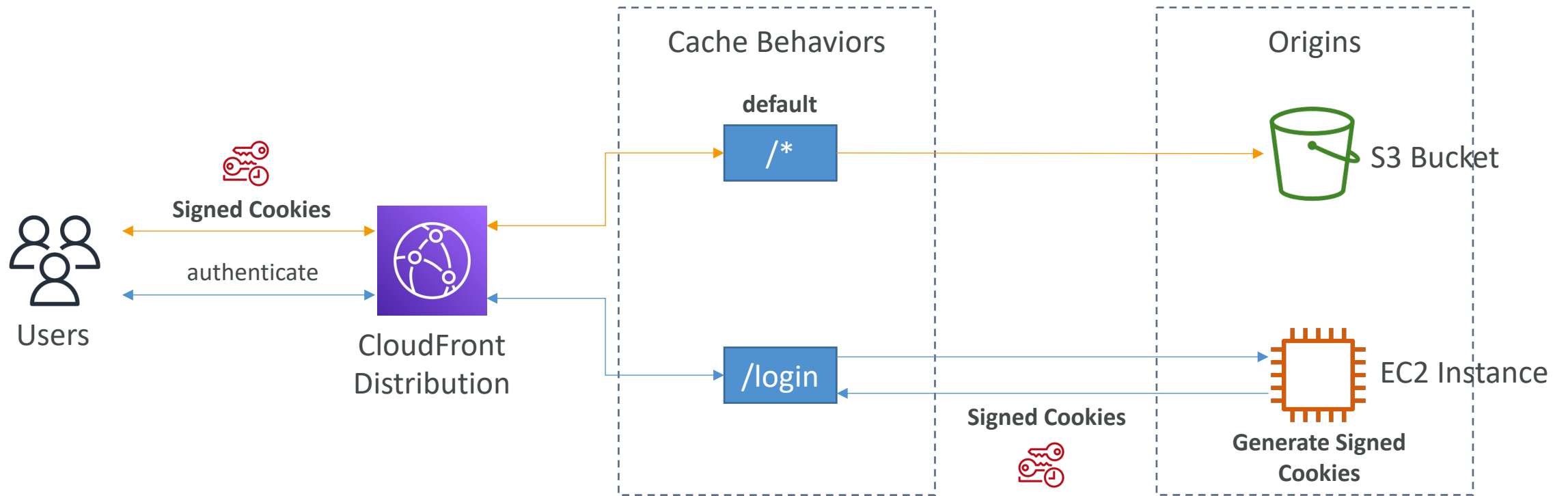


CloudFront – Cache Behaviors

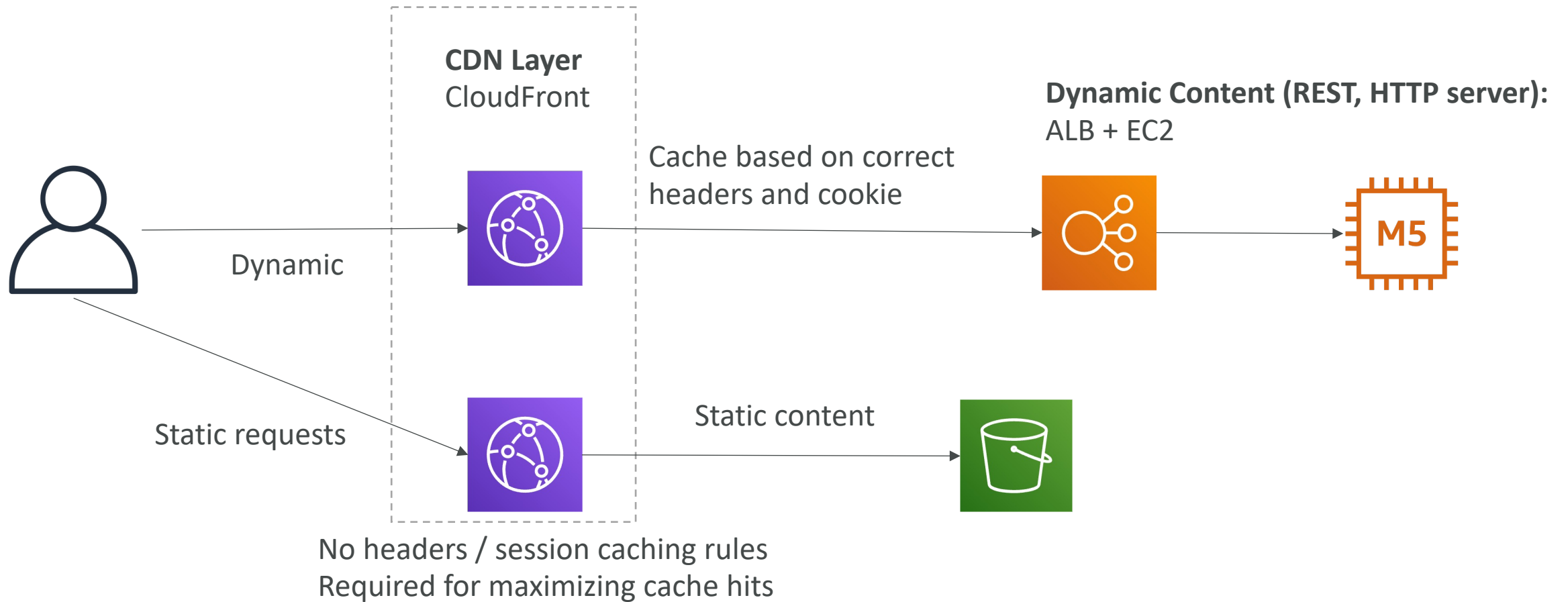
- Configure different settings for a given URL path pattern
- Example: one specific cache behavior to `images/*.jpg` files on your origin web server
- Route to different kind of origins/origin groups based on the content type or path pattern
 - `/images/`
 - `/api/`
 - `/*` (default cache behavior)
- When adding additional Cache Behaviors, the Default Cache Behavior is always the last to be processed **and is always `/*`**



CloudFront – Cache Behaviors – Sign In Page

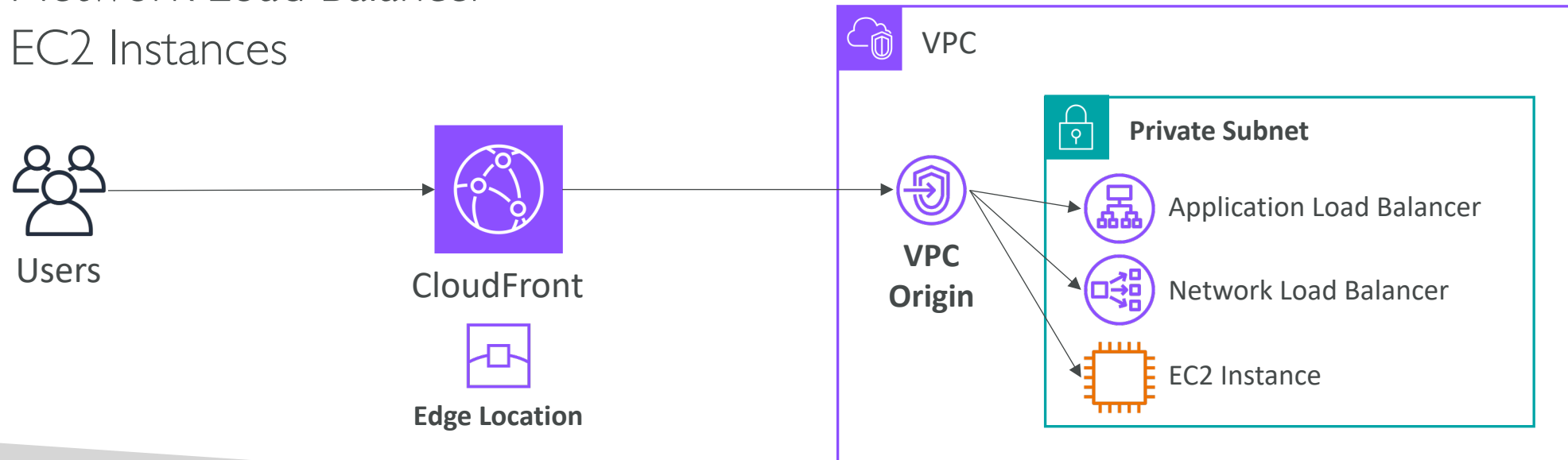


CloudFront – Maximize cache hits by separating static and dynamic distributions

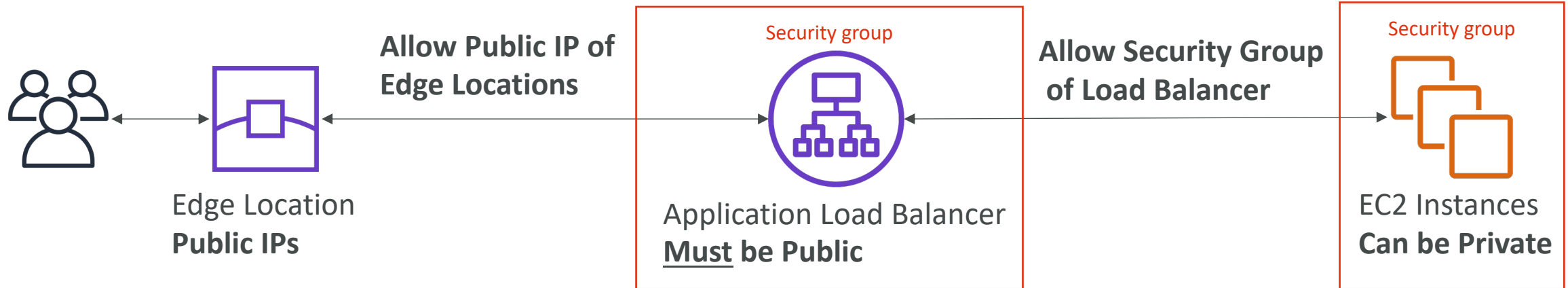
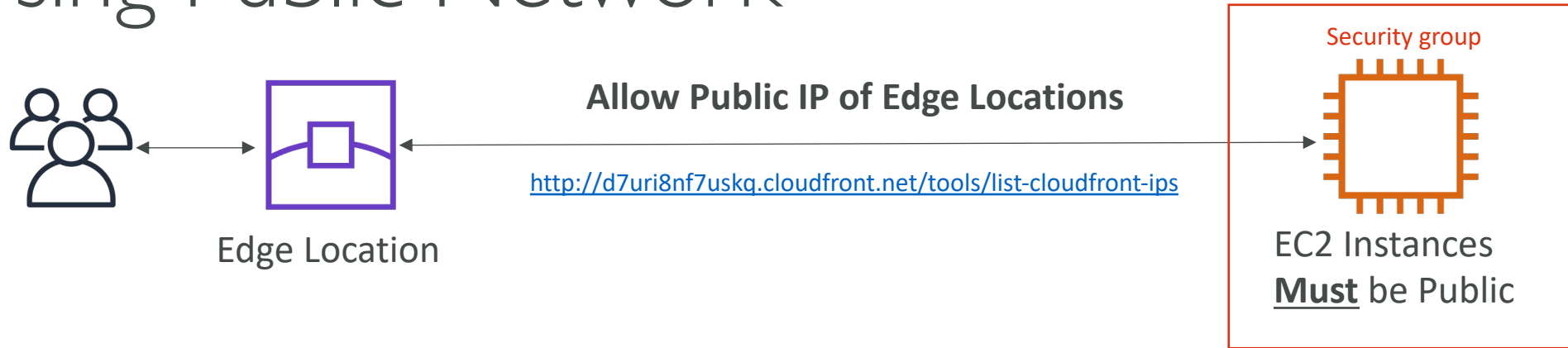


CloudFront – ALB or EC2 as an origin Using VPC Origins

- Allows you to deliver content from your applications hosted in your VPC private subnets (no need to expose them on the Internet)
- Deliver traffic to **private**:
 - Application Load Balancer
 - Network Load Balancer
 - EC2 Instances



CloudFront – ALB or EC2 as an origin Using Public Network



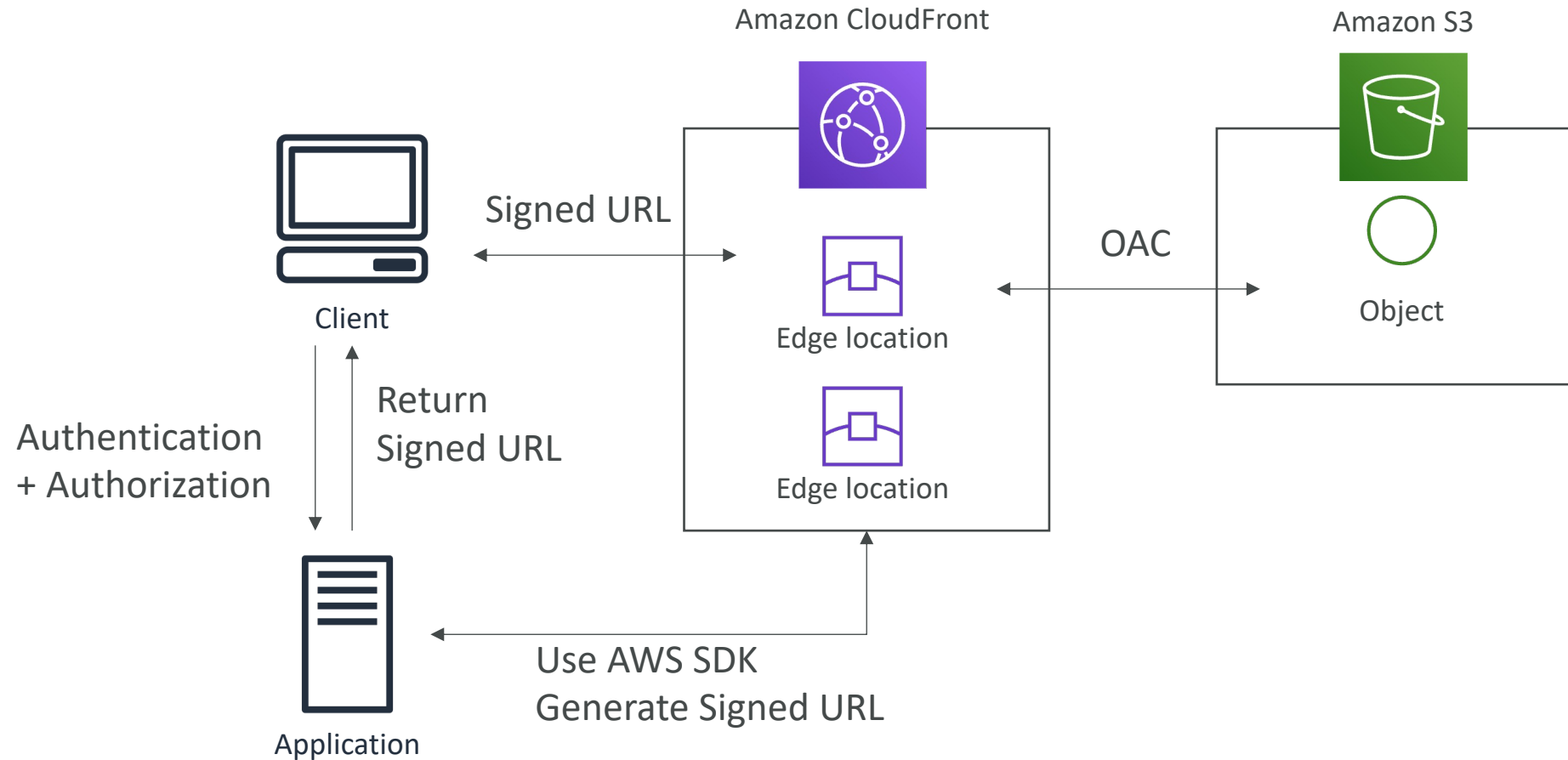
CloudFront Geo Restriction

- You can restrict who can access your distribution
 - **Allowlist:** Allow your users to access your content only if they're in one of the countries on a list of approved countries.
 - **Blocklist:** Prevent your users from accessing your content if they're in one of the countries on a list of banned countries.
- The “country” is determined using a 3rd party Geo-IP database
- Use case: Copyright Laws to control access to content

CloudFront Signed URL / Signed Cookies

- You want to distribute paid shared content to premium users over the world
- We can use CloudFront Signed URL / Cookie. We attach a policy with:
 - Includes URL expiration
 - Includes IP ranges to access the data from
 - Trusted signers (which AWS accounts can create signed URLs)
- How long should the URL be valid for?
 - Shared content (movie, music): make it short (a few minutes)
 - Private content (private to the user): you can make it last for years
- Signed URL = access to individual files (one signed URL per file)
- Signed Cookies = access to multiple files (one signed cookie for many files)

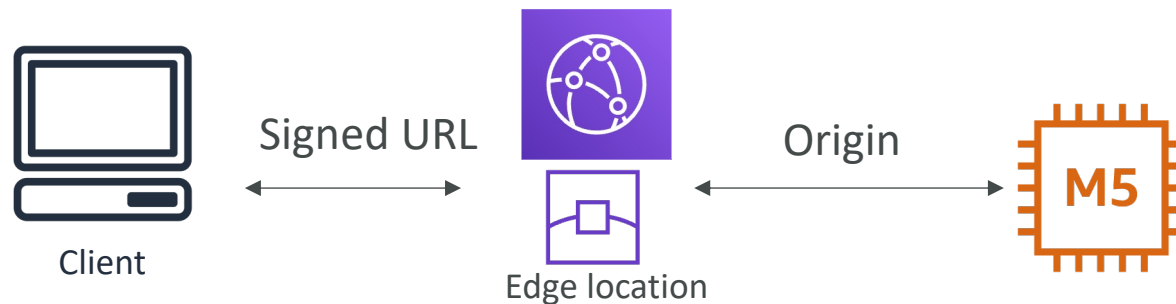
CloudFront Signed URL Diagram



CloudFront Signed URL vs S3 Pre-Signed URL

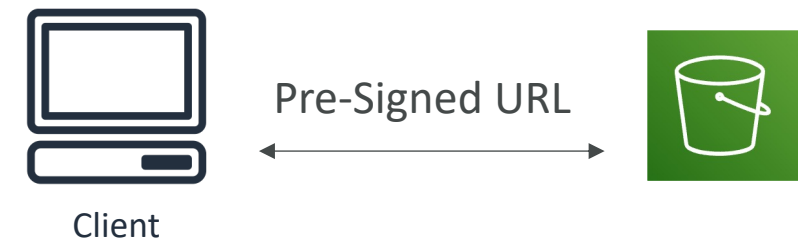
- CloudFront Signed URL:

- Allow access to a path, no matter the origin
- Account wide key-pair, only the root can manage it
- Can filter by IP, path, date, expiration
- Can leverage caching features



- S3 Pre-Signed URL:

- Issue a request as the person who pre-signed the URL
- Uses the IAM key of the signing IAM principal
- Limited lifetime



CloudFront Signed URL Process

- Two types of signers:
 - Either a trusted key group (recommended)
 - Can leverage APIs to create and rotate keys (and IAM for API security)
 - An AWS Account that contains a CloudFront Key Pair
 - Need to manage keys using **the root account and the AWS console**
 - Not recommended because you shouldn't use the root account for this
- In your CloudFront distribution, create one or more **trusted key groups**
- You generate your own public / private key
 - The private key is used by your applications (e.g. EC2) to sign URLs
 - The public key (uploaded) is used by CloudFront to verify URLs

CloudFront - Pricing

- CloudFront Edge locations are all around the world
- The cost of data out per edge location varies

Per Month	United States, Mexico, & Canada	Europe & Israel	South Africa, Kenya, & Middle East	South America	Japan	Australia & New Zealand	Hong Kong, Philippines, Singapore, South Korea, Taiwan, & Thailand	India
First 10TB	\$0.085	\$0.085	\$0.110	\$0.110	\$0.114	\$0.114	\$0.140	\$0.170
Next 40TB	\$0.080	\$0.080	\$0.105	\$0.105	\$0.089	\$0.098	\$0.135	\$0.130
Next 100TB	\$0.060	\$0.060	\$0.090	\$0.090	\$0.086	\$0.094	\$0.120	\$0.110
Next 350TB	\$0.040	\$0.040	\$0.080	\$0.080	\$0.084	\$0.092	\$0.100	\$0.100
Next 524TB	\$0.030	\$0.030	\$0.060	\$0.060	\$0.080	\$0.090	\$0.080	\$0.100
Next 4PB	\$0.025	\$0.025	\$0.050	\$0.050	\$0.070	\$0.085	\$0.070	\$0.100
Over 5PB	\$0.020	\$0.020	\$0.040	\$0.040	\$0.060	\$0.080	\$0.060	\$0.100

lower

higher

CloudFront – Price Classes

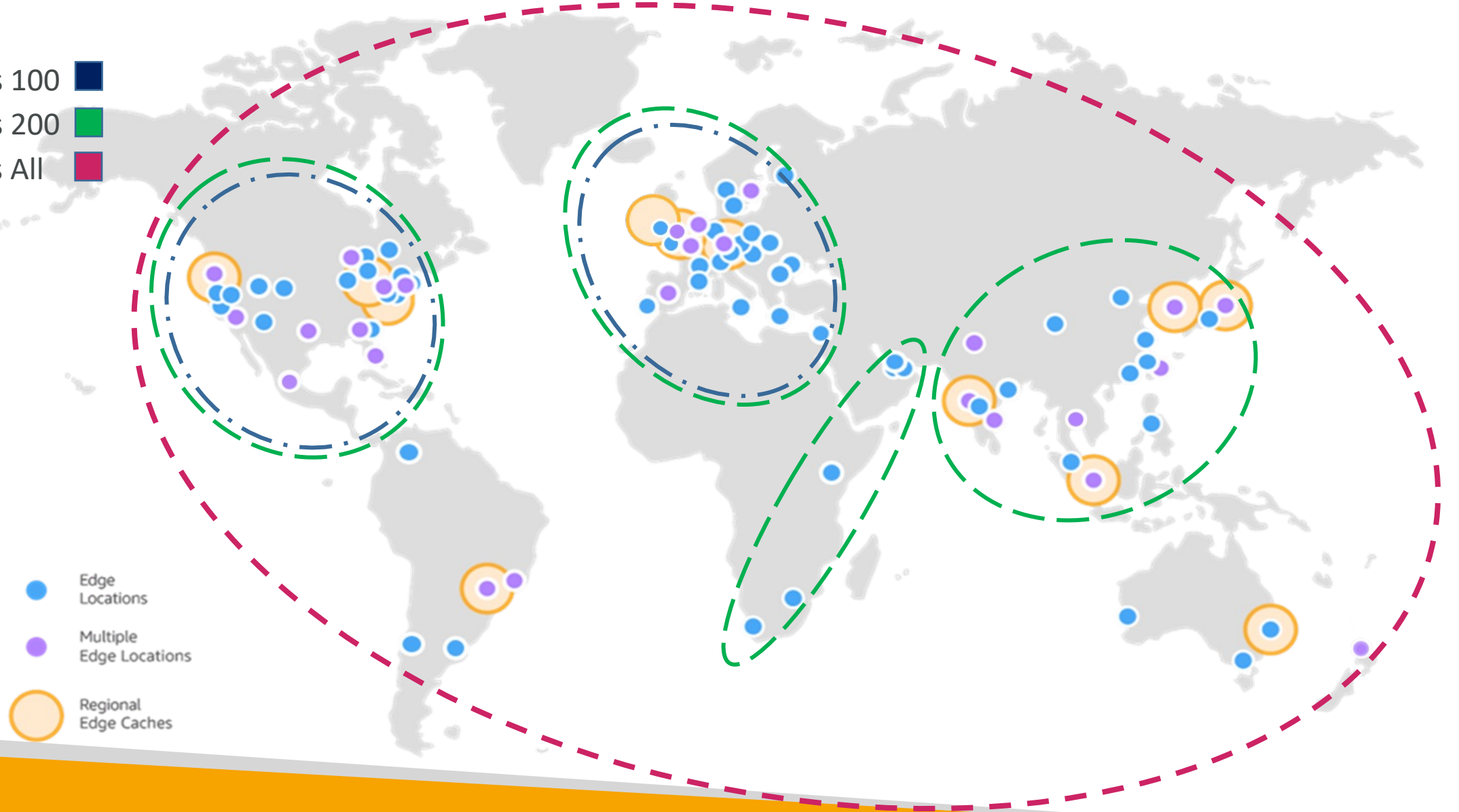
- You can reduce the number of edge locations for **cost reduction**
- Three price classes:
 1. Price Class All: all regions – best performance
 2. Price Class 200: most regions, but excludes the most expensive regions
 3. Price Class 100: only the least expensive regions

Edge Locations Included Within	United States, Mexico, & Canada	Europe & Israel	South Africa, Kenya, & Middle East	South America	Japan	Australia & New Zealand	Hong Kong, Philippines, Singapore, South Korea, Taiwan, & Thailand	India
Price Class All	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Price Class 200	Yes	Yes	Yes	x	Yes	x	Yes	Yes
Price Class 100	Yes	Yes	x	x	x	x	x	x

CloudFront - Price Class

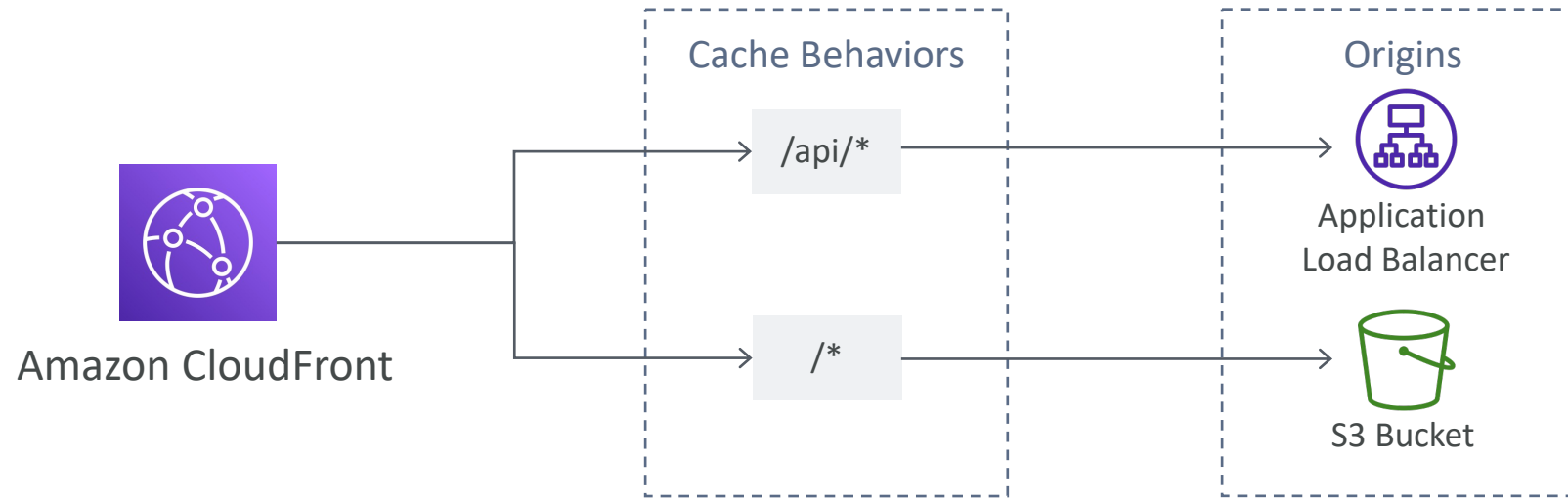
Prices Class 100 ■
Prices Class 200 ■
Prices Class All ■

● Edge Locations
● Multiple Edge Locations
○ Regional Edge Caches



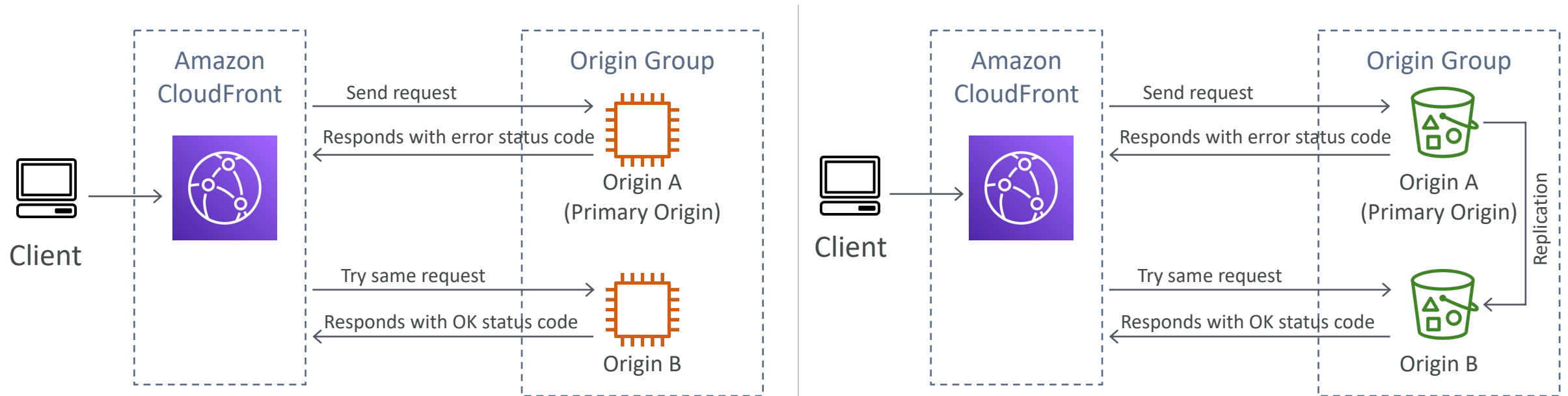
CloudFront – Multiple Origin

- To route to different kind of origins based on the content type
- Based on path pattern:
 - /images/*
 - /api/*
 - /*



CloudFront – Origin Groups

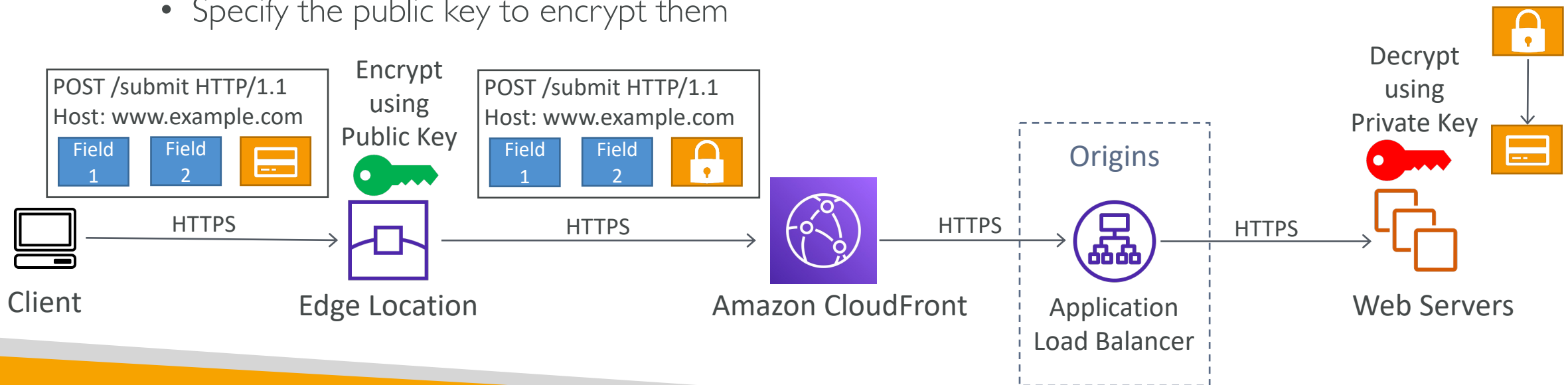
- To increase high-availability and do failover
- Origin Group: one primary and one secondary origin
- If the primary origin fails, the second one is used



S3 + CloudFront – Region-level High Availability

CloudFront – Field Level Encryption

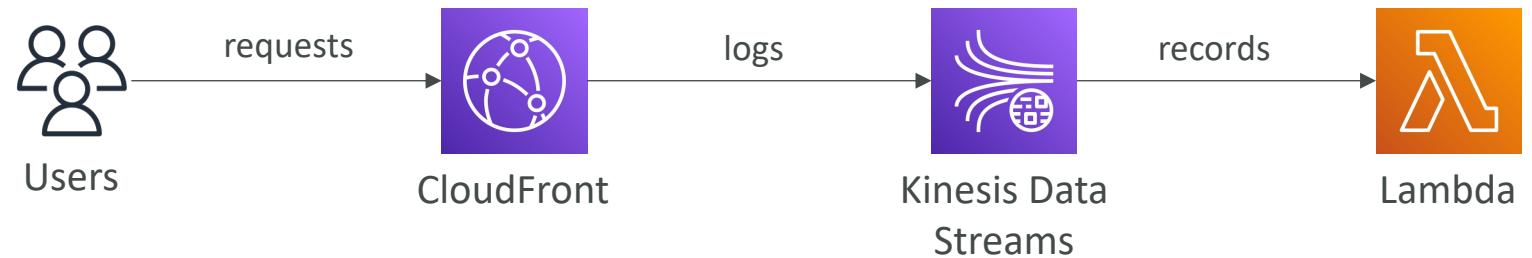
- Protect user sensitive information through application stack
- Adds an additional layer of security along with HTTPS
- Sensitive information encrypted at the edge close to user
- Uses asymmetric encryption
- Usage:
 - Specify set of fields in POST requests that you want to be encrypted (up to 10 fields)
 - Specify the public key to encrypt them



CloudFront – Real Time Logs

- Get real-time requests received by CloudFront sent to Kinesis Data Streams
- Monitor, analyze, and take actions based on content delivery performance
- Allows you to choose:
 - **Sampling Rate** – percentage of requests for which you want to receive
 - Specific fields and specific Cache Behaviors (path patterns)

Real-time Processing



Near Real-time Processing

